

A SHORT NOTE ON THE TRANSFINITE DIAMETER OF HEDGEHOGS WITH 2 QUILLS

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ABSTRACT. We produce an effective method to compute the transfinite diameter of a hedgehog generated by two complex numbers and provide a lower bound for it improving Dubinin's one.

1. INTRODUCTION

Two complex numbers $z, w \in \mathbb{C}$ delimit the line segment

$$[z, w] = \{(1 - t)z + tw : 0 \leq t \leq 1\}.$$

The *hedgehog* generated by $z_1, \dots, z_n \in \mathbb{C}$ is defined as

$$H(z_1, \dots, z_n) = \bigcup_{i=1}^n [0, z_i].$$

The segments $[0, z_i]$ are called *quills*. In this note, we find a formula to compute the transfinite diameter of hedgehogs generated by pairs of complex numbers.

For a compact set $K \subseteq \mathbb{C}$, define the sequence $(d_n(K))_{n \geq 2}$ by

$$d_n(K) = \sup_{w_1, \dots, w_n \in K} \prod_{1 \leq i < j \leq n} |w_i - w_j|^{\frac{2}{n(n-1)}}.$$

This sequence is monotonically decreasing [Ran95, Theorem 5.5.2]. The *transfinite diameter* of K is defined as the limit of $d_n(K)$ and is denoted by $d(K)$. Fekete and Szegő proved that the transfinite diameter of compact subsets of $\mathbb{C}$ equals their potential-theoretic capacity; see [Ran95, Ch. 5]. It follows from the definition that the transfinite diameter of a set is invariant under isometries. Moreover, if $\lambda K = \{\lambda z : z \in K\}$, then $d(\lambda K) = |\lambda| \cdot d(K)$. Computing the transfinite diameter of a compact set is generally a difficult problem; a few cases for which this computation is known are listed in [Ran95, Table 5.1].

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In the case of hedgehogs, no explicit formulas are known. The best result is the following estimate.

Theorem 1.1 (Dubinin, [Dub84]). *Let $z_1, \dots, z_n \in \mathbb{C}$. Then*

$$d(H(z_1, \dots, z_n)) \leq 4^{-\frac{1}{n}} \max_{1 \leq i \leq n} |z_i|.$$

Moreover, equality holds when the quills are equally spaced in angle.

In this note we describe a method that allows to numerically compute the transfinite diameter of a 2-hedgehog and derive an estimate that improves [Theorem 1.1](#) for $n = 2$.

2. RIEMANN MAPPING

Denote by $\mathbb{C}_\infty$ the Riemann sphere and by D the unit disk. For a hedgehog H with $d(H) = 1$, Schmidt computed the Riemann map explicitly.

Proposition 2.1 (Schmidt [Sch]). *Let H be a hedgehog generated by n complex numbers. If $d(H) = 1$, then there exist $w_1, \dots, w_n \in \partial D$ and $p_1, \dots, p_n \in [0, 1]$, with $p_1 + \dots + p_n = 2$, such that the map*

$$\begin{aligned} \Psi: \mathbb{C}_\infty \setminus D &\longrightarrow \mathbb{C}_\infty \setminus H \\ z &\longmapsto C \prod_{i=1}^n \left((z - w_i) \left(\frac{1}{z} - \bar{w}_i \right) \right)^{p_i/2} \end{aligned}$$

is a Riemann map, where

$$C = \prod_{i=1}^n (-\bar{w}_i)^{-p_i/2}.$$

In the above proposition, the branches of the fractional powers are chosen so that $\Psi(z) = z + O(1)$ as $z \rightarrow \infty$.

We assume that the quills $[0, z_i]$ are ordered counterclockwise. It follows from Schmidt's proof that the points w_i are the boundary preimages of 0 under the continuous extension of Ψ to ∂D and $p_i = \vartheta_i/\pi$ where ϑ_i is the oriented angle between the quill i and $i + 1$, with indices understood modulo n . Such a continuous extension maps the counterclockwise arc between w_i and w_{i+1} onto the quill $[0, z_i]$.

For $n = 2$, after rotation we may assume that $w_1 = 1$, and write $w_2 = e^{is}$ for some $s \in [0, \pi]$. If $p = \vartheta/\pi$, then the Riemann parameterization is

$$(1) \quad \Psi(e^{it}) = C \left((e^{it} - 1) (e^{-it} - 1) \right)^{p/2} \left((e^{it} - e^{is}) (e^{-it} - e^{-is}) \right)^{1-p/2}$$

where $C = (-1)^{-p/2} (-e^{-is})^{-1+p/2}$.

For fixed ϑ , the values of s parameterize all possible 2-hedgehogs of angle ϑ and transfinite diameter 1. Given a 2-hedgehog H , we call the corresponding parameter s the *parameter of H* .

We will mainly focus on the absolute value

$$f(t) = |\Psi(e^{it})| = 2 (1 - \cos t)^{p/2} (1 - \cos(t - s))^{1-p/2}.$$

The parameter $2\pi - s$ corresponds to the hedgehog for which the two quills are switched.

Remark 2.2. To simplify the notation, we write $\sin^n(x) = (\sin x)^n$.

3. PARAMETERIZING 2-HEDGEHOGS

Let H be a 2-hedgehog of transfinite diameter 1. Up to rotation, for the rest of this note we assume that H is generated by z_r and z_c with z_r a positive real number and z_c a complex number with positive imaginary part. The counterclockwise angle from $[0, z_r]$ to $[0, z_c]$ is denoted by ϑ . Since there exists a formula for the transfinite diameter of the segment — namely, one fourth of the segment length — we exclude the degenerate cases $\vartheta \in \{0, \pi\}$. Recall that s denotes the parameter of H . Up to symmetry and up to discarding the cases with $z_r = 0$ or $z_c = 0$, we may assume $s \in (0, \pi)$.

The goal of this section is to show that 2-hedgehogs of fixed angle are parametrized by the ratio $|z_r|/|z_c|$.

Proposition 3.1. *Fix $p \in (0, 1)$. The map*

$$\begin{aligned} h: \mathcal{H}_\vartheta &\longrightarrow [0, 1) \\ H(z_r, z_c) &\longmapsto \frac{|z_r|}{|z_c|}. \end{aligned}$$

is a bijection from the set $\mathcal{H}_\vartheta$ of 2-hedgehogs with angle $\vartheta = p\pi$ and $|z_r| < |z_c|$ onto $[0, 1)$.

Lemma 3.2. *With the above notation we have $z_r = f(t_r)$ and $z_c = f(t_c)e^{i\vartheta}$ where*

$$t_r = \frac{s}{2} + \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right)$$

and

$$t_c = \frac{s}{2} + \pi - \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right)$$

Remark 3.3. We denote by $\arcsin: [-1, 1] \rightarrow [-\pi/2, \pi/2]$ the principal value of the arcsine.

Proof. Recall that

$$f(t) = 2(1 - \cos t)^{p/2}(1 - \cos(t-s))^{1-p/2}.$$

For $t \in (0, 2\pi) \setminus \{s\}$, we compute

$$f'(t) = \frac{4 \left((p-1) \sin \left(\frac{s}{2} \right) + \sin \left(\frac{s}{2} - t \right) \right) \sin \left(\frac{s-t}{2} \right) \sin \left(\frac{t}{2} \right)}{(1 - \cos(s-t))^{p/2} (1 - \cos(t))^{1-p/2}}.$$

Hence $f'(t) = 0$ if and only if

$$(2) \quad \sin \left(t - \frac{s}{2} \right) = (p-1) \sin \left(\frac{s}{2} \right).$$

Since $p \in (0, 1)$ and $s \in (0, \pi)$, the right-hand side of (2) lies in $(-1, 0)$. Therefore, the solutions of (2) for $t \in (0, 2\pi) \setminus \{s\}$ are

$$t_r = \frac{s}{2} + \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right)$$

and

$$t_c = \frac{s}{2} + \pi - \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right).$$

It remains to identify them as the endpoints of the two quills. Since $f(0) = f(s) = 0$ and $f(t) > 0$ for $t \in (0, 2\pi) \setminus \{s\}$, each of the intervals $(0, s)$ and $(s, 2\pi)$ contains at least one local maximum. Because f has only two critical points in $[0, 2\pi)$, namely t_r and t_c , these are the unique critical points on the two intervals, hence the corresponding local maxima. This proves the lemma. ■

Lemma 3.4. *If $0 < s < \pi$, then $t_r < t_c$ and $f(t_r) < f(t_c)$.*

Proof. For the first claim, it is enough to note that

$$t_r - t_c = 2 \arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right) - \pi$$

is negative, as $p-1 \in (-1, 0)$ and $s \in (0, 2\pi)$ imply

$$\arcsin \left((p-1) \sin \left(\frac{s}{2} \right) \right) < \frac{\pi}{2}.$$

We now prove that $f(t_r) < f(t_c)$. The identity $1 - \cos x = 2 \sin^2(x/2)$ implies

$$f(t_r) = 4 \sin^p \left(\frac{t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{s - t_r}{2} \right).$$

Use the identity $t_c = s + \pi - t_r$ to obtain

$$f(t_c) = 4 \sin^p \left(\frac{s + \pi - t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{t_r - \pi}{2} \right) = 4 \cos^p \left(\frac{s - t_r}{2} \right) \cdot \cos^{2-p} \left(\frac{t_r}{2} \right).$$

Since the sine function is increasing in $(0, \pi/2)$ and

$$\frac{t_r}{2} + \frac{s - t_r}{2} = \frac{s}{2} < \frac{\pi}{2},$$

we conclude that

$$\sin \frac{t_r}{2} \leq \sin \left(\frac{\pi}{2} - \frac{s - t_r}{2} \right) = \cos \frac{s - t_r}{2}.$$

The inequality $\sin((s - t_r)/2) < \cos(t_r/2)$ is derived similarly. Therefore, the inequality $f(t_r) < f(t_c)$ follows. $\blacksquare$

The idea behind the proof of **Theorem 3.1** is that 2-hedgehogs of fixed angle are characterized by the ratio $f(t_r)/f(t_c)$.

Proof of Theorem 3.1. Let H be the hedgehog generated by z_r and z_c . Let s be the Riemann parameter of H . **Theorem 3.2** implies that $z_r = f(t_r)$ and $z_c = f(t_c)e^{i\vartheta}$. Up to replacing s with $2\pi - s$, we may assume $s \in [0, \pi]$.

Consider t_r and t_c as functions of s . We prove that the function

$$h(s) = \frac{f(t_r(s))}{f(t_c(s))}$$

is a bijection. This function can be written as

$$h(s) = g_+(\tilde{t}_r(s)) \cdot \frac{1}{g_-(\tilde{t}_c(s))}$$

where

$$g_{\pm}(t) = \frac{(1 - \cos^2 t)^p}{1 \pm \cos t},$$

$$\tilde{t}_r(s) = \frac{s}{2} + \arcsin\left((p-1)\sin\frac{s}{2}\right)$$

and

$$\tilde{t}_c(s) = \frac{s}{2} - \arcsin\left((p-1)\sin\frac{s}{2}\right).$$

The functions g_+ , $1/g_-$, $\tilde{t}_r$ and $\tilde{t}_c$ are non-negative and monotonically increasing in t and s respectively. Hence, h is monotonically increasing and continuous. Since

$$h(0) = 0 \quad \text{and} \quad \lim_{s \rightarrow \pi} h(s) = 1,$$

the function h is a bijection from $[0, \pi]$ onto $[0, 1]$. ■

4. TRANSFINITE DIAMETER OF 2-HEDGEHOGS

In this section, we derive a formula for the transfinite diameter of a 2-hedgehog and prove a bound depending only on the quill lengths.

4.1. Explicit computation of the transfinite diameter. Let H be a 2-hedgehog. Denote the endpoints of the quills by Z_r and Z_c . Up to rotation and symmetries, we may assume that Z_r is a positive real number, that $|Z_r| \leq |Z_c|$ and that Z_c has non-negative imaginary part. Define $z_r = Z_r/d(H)$ and $z_c = Z_c/d(H)$. Then, the hedgehog $\tilde{H} = H(z_r, z_c)$ has transfinite diameter 1. The Riemann parameter of $\tilde{H}$ is

$$\tilde{s} = h^{-1}\left(\frac{|z_r|}{|z_c|}\right) = h^{-1}\left(\frac{|Z_r|}{|Z_c|}\right).$$

We conclude that

$$d(H) = \frac{Z_r}{z_r} = \frac{Z_r}{f(t_r(\tilde{s}))} = \frac{Z_r}{f\left(t_r\left(h^{-1}\left(\frac{|Z_r|}{|Z_c|}\right)\right)\right)}.$$

4.2. A Dubinin-type estimate.

Proposition 4.1. *Let $H = H(Z_r, Z_c)$ be a 2-hedgehog. Then*

$$d(H) \leq \frac{|Z_r| + |Z_c|}{4}.$$

Proof. Let z_r and z_c be as in the previous section. It suffices to show that $|z_r| + |z_c| \geq 4$.

To simplify the computation, write

$$f(t) = 4 \sin^p \left(\frac{t}{2} \right) \sin^{2-p} \left(\frac{t-s}{2} \right).$$

It follows that

$$\begin{aligned} |z_r| + |z_c| &= f(t_r) + f(t_c) \\ &= 4 \sin^p \left(\frac{t_r}{2} \right) \cdot \sin^{2-p} \left(\frac{s-t_r}{2} \right) + 4 \sin^p \left(\frac{t_c}{2} \right) \cdot \sin^{2-p} \left(\frac{s-t_c}{2} \right). \end{aligned}$$

Use that $t_c = s + \pi - t_r$ to write

$$(3) \quad |z_r| + |z_c| = 4 \sin^p \left(\frac{t_r}{2} \right) \sin^{2-p} \left(\frac{s-t_r}{2} \right) + 4 \cos^p \left(\frac{s-t_r}{2} \right) \cos^{2-p} \left(\frac{t_r}{2} \right).$$

Note that t_r fullfills (2). Rewrite it as

$$(4) \quad \sin \left(\frac{t}{2} + \left(\frac{t}{2} - \frac{s}{2} \right) \right) = (p-1) \sin \left(\frac{t}{2} - \left(\frac{t}{2} - \frac{s}{2} \right) \right).$$

The standard trigonometric identities imply that

$$\sin \left(\frac{t}{2} + \left(\frac{t}{2} - \frac{s}{2} \right) \right) = \sin \frac{t}{2} \cos \left(\frac{t}{2} - \frac{s}{2} \right) + \cos \frac{t}{2} \sin \left(\frac{t}{2} - \frac{s}{2} \right)$$

and that

$$\sin \left(\frac{t}{2} - \left(\frac{t}{2} - \frac{s}{2} \right) \right) = \sin \frac{t}{2} \cos \left(\frac{t}{2} - \frac{s}{2} \right) - \cos \frac{t}{2} \sin \left(\frac{t}{2} - \frac{s}{2} \right).$$

Substitute these last two identities in (4) and rearrange the expression to get the identity

$$(5) \quad p \cot \left(\frac{t_r}{2} \right) = (2-p) \cot \left(\frac{s-t_r}{2} \right)$$

Now we prove that (3) is decreasing in p . Consider first

$$F(p) = \sin^p \left(\frac{t_r}{2} \right) \sin^{2-p} \left(\frac{s-t_r}{2} \right).$$

The function $F(p)$ is positive. A simple computation shows that

$$\frac{d \log F(p)}{dp} = \log \left(\frac{\sin \frac{t_r}{2}}{\sin \frac{s-t_r}{2}} \right) + \frac{t_r}{2} \left(p \cot \left(\frac{t_r}{2} \right) - (2-p) \cot \left(\frac{s-t_r}{2} \right) \right)$$

which, due to (5) yields to

$$\frac{d \log F(p)}{dp} = \log \left(\sin \frac{t_r}{2} \right) - \log \left(\sin \frac{s - t_r}{2} \right)$$

Since $0 \leq t_r/2 \leq (s - t_r)/2 < \pi/2$ and the sine is strictly increasing in $(0, \pi/2)$, we conclude that the logarithmic derivative is non-positive. Note that

$$\frac{dF(p)}{dp} = F(p) \cdot \frac{d \log F(p)}{dp} \leq 0,$$

hence we conclude that F is non-increasing in p .

Consider now

$$G(p) = \cos^p \left(\frac{s - t_r}{2} \right) \cos^{2-p} \left(\frac{t_r}{2} \right).$$

The function $G(p)$ is positive. A similar computation as for F yields

$$\frac{d \log G(p)}{dp} = \log \left(\cos \frac{s - t_r}{2} \right) - \log \left(\cos \frac{t_r}{2} \right).$$

Since $0 \leq t_r/2 \leq (s - t_r)/2 < \pi/2$ and the cosine is strictly decreasing in $(0, \pi/2)$, we conclude that

$$\frac{dG(p)}{dp} = G(p) \cdot \frac{d \log G(p)}{dp} \leq 0,$$

hence we conclude that G is non-increasing in p .

It follows that (3) is non-increasing in p , from which

$$|z_r| + |z_c| = 4F(p) + 4G(p) \geq 4F(1) + 4G(1) = 4 \sin^2 \left(\frac{s}{4} \right) + 4 \cos^2 \left(\frac{s}{4} \right) = 4.$$

The proof follows. ■

Remark 4.2. The bound is sharp. It is attained by the hedgehogs generated by Z_r and $Z_c = -Z_r$.

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